Nayoung Choi

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Research Interests

Natural Language Processing (NLP) / Information Retrieval (IR) / Explainable Artificial Intelligence (XAI)

Educations

M.S. **Yonsei University** Seoul, South Korea
- Department of Digital Analytics / Faculty of Artificial Intelligence Convergence 2020-03 ~ 2022-02

GPA: 4.25 out of 4.3 (On a 4.3 scale)
Relevant Courses: Computer Programming / Database Management / Big Data Parallel Processing / Linear Algebra / Statistical Analysis of Big Data / Machine Learning / Artificial Intelligence and Deep Learning / Text Mining / Natural Language Processing and Deep Learning / Recommender System

B.A. **Sungkyunkwan University** Seoul, South Korea
- Department of Library and Information Science / Faculty of Liberal Arts 2013-03 ~ 2018-02

GPA: 3.7 out of 4.5 (On a 4.5 scale)
Relevant Courses: Introduction to Information Science / Introduction to Linguistics / Introduction to Information Organization / Information Organization I / Information Analysis and Evaluation / Introduction to Data Science

\- **Vilnius University** Vilnius, Lithuania
- Exchange student for Fall 2016 2016-08 ~ 2017-01

Research Experiences

DeepText Lab (Graduate Student Researcher) 2020-03 ~ 2022-02
- Advisor: Min Song https://deeptext.yonsei.ac.kr

EmoryNLP (Collaborative Researcher) 2020-08 ~ 2021-08
- Advisor: Jinho D. Choi https://www.emorynlp.org/projects/emory-yonsei-collaboration

Industrial Experiences

NAVER Corporation (AI/ML Engineer, for search engine)
- **Full-time** 2022-01 ~ Present
  - Made a part of safety module for conversational search engine, to detect and properly respond to toxic questions
  - Conducted experiments to classify the theme of search queries using our in-house generative LLM, HyperCLOVA
  - Developed a model to detect low-quality documents based on implicit user feedback to re-rank search engine results
- **Internship** 2021-08 ~ 2021-11
  - Improved search engine results for ambiguous queries by enhancing document representations towards Word Sense Disambiguation using pre-trained language models

Publications

[1] Nayoung Choi. **Breaking Down Word Semantics from Pre-trained Language Models through Layer-wise Dimension Selection.** *(submitted to ACL Rolling Review)* [arxiv]

[2] I. Paek, N. Choi, S. Ha, Y. Kim and M. Song, "**SQ2SV: Sequential Queries to Sequential Videos retrieval,**" *2022 IEEE International Conference on Big Data (Big Data),* Osaka, Japan, 2022, pp. 3631-3634, doi: 10.1109/BigData55660.2022.10020744. [paper] [presentation]

[3] Sooyoun Han, Sumin Seo, Minji Kang, Jongin Kim, Nayoung Choi, Min Song, and Jinho D. Choi. 2021. **FantasyCoref: Coreference Resolution on Fantasy Literature Through Omniscient Writer's Point of View.** *In Proceedings of the Fourth Workshop on Computational Models of Reference, Anaphora and Coreference,* pages 24–35, Punta Cana, Dominican Republic. Association for Computational Linguistics. [paper] [data]

[4] Jongin Kim, Nayoung Choi, Seunghyun Lim, Jungwhan Kim, Soojin Chung, Hyunsoo Woo, Min Song, and Jinho D. Choi. 2021. **Analysis of Zero-Shot Crosslingual Learning between English and Korean for Named Entity Recognition.** *In Proceedings of the 1st Workshop on Multilingual Representation Learning,* pages 224–237, Punta Cana, Dominican Republic. Association for Computational Linguistics. [paper] [code] [data]

Awards

2021 NAVER AI RUSH (3rd place) NAVER Corp., South Korea
- Developed a grammar error correction model by enhancing Transformer-based sequence-to-sequence architecture with multi-task learning, incorporating additional CNN layers to predict grammar error types together [code]

2020 Data Preprocessing Competition (1st place) Dept. of Digital Analytics, Yonsei University, South Korea

- Participated in the on-campus competition, in which all department students participated, and solved the given three data preprocessing problems

2020 AI Problem Solving Contest (3rd place) National IT Industry Promotion Agency, South Korea

- Developed a time-series indoor air quality prediction model using LSTM layers

2020 Big Contest (2nd place) Ministry of Science and ICT, South Korea

- Developed an optimal broadcast scheduling algorithm for a television shopping channel using bipartite matching, leveraging LightGBM-based sales performance prediction

2019 Whether Bigdata Contest (selected in preliminary round) Meteorological Administration, South Korea

- Proposed an optimization strategy for identifying optimal sales periods for drug stores by analyzing public weather and distribution industry data

Projects

* The projects below were undertaken as industry-academic collaborations between my affiliated organization, Yonsei University, and Corporate Partners from the industry.

Analyzing social media posts about the Seoul mayoral election for Channel A News 2021-03 ~ 2021-04

- Collected social media posts and performed sentiment analysis using word2vec on content relevant to the two mayoral candidates [news video]

Building an internal data management system for JTBC Dramahouse Corporation 2020-10 ~ 2021-04

- Built a web-based system to manage various types of data related to drama production, using Python Django framework for development and AWS platform for deployment

Analyzing the association between intraoperative hemodynamic parameter changes and acute kidney injury after heart valve surgery for Yonsei Severance Hospital 2020-02 ~ 2021-02

- Conducted a study investigating factors involved in the occurrence of acute kidney injury after heart valve surgery by analyzing time-series vital sign data during surgery in patients

Others

Educational Volunteer Services

Seocho Children's Center Seoul, South Korea

- Taught Mathematics to the center children for a semester Fall 2017

Samgaksan High School Seoul, South Korea

- Taught Korean language to high school students for a semester Fall 2017

Scholarships

YulGok Scholarship Sungkyunkwan University

- Received a 50% tuition fee scholarship for four years due to my high Korean SAT scores Spring 2013 to Fall 2017

Assistant Scholarship Yonsei University

- Deployed and tested a personal information de-identification solution assisting Professor Wonseok Lee Fall 2020

Programming Skill Sets

Language – Python / C++ / Scala / Java
MLOps – Apache Airflow / Kubeflow / Spark
Web Framework – Python Django, Flask
Database – SQL / HQL